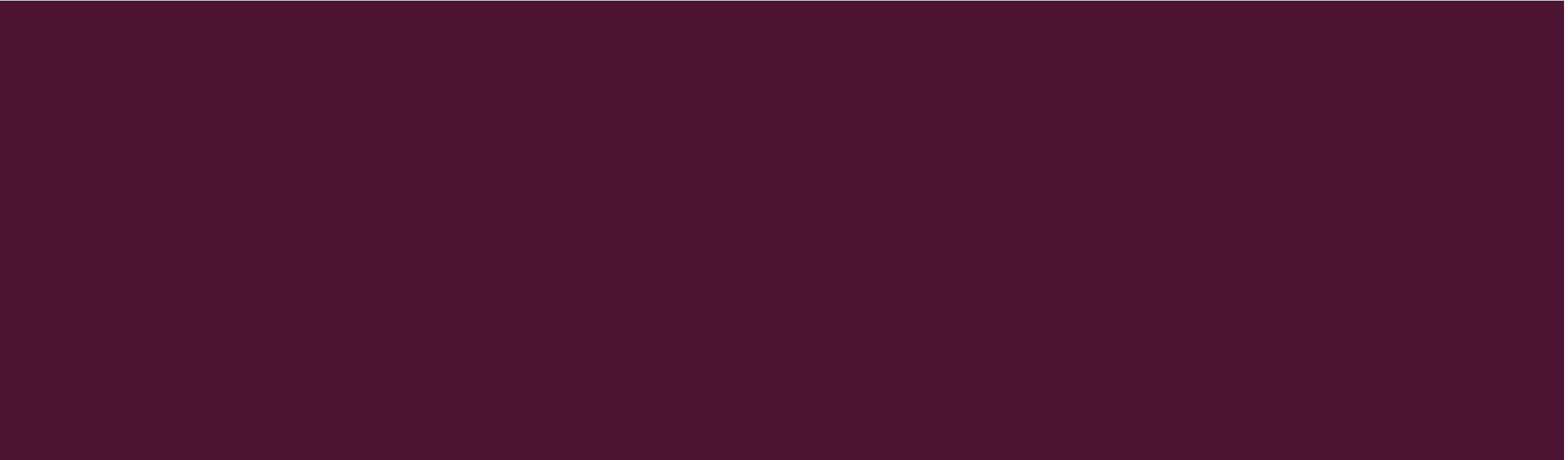




ORDER OF VOLATILITY



FORENSIC METHODOLOGIES

- **Traditional Forensics**

- Analyzing a “dead” system that has had its power cord pulled
- Least chance of modifying data on disk, but “live” data is lost forever

- **Live Forensics**

- Methodology which advocates extracting “live” system data before pulling the cord to preserve memory, process, and network information
- Goal is to minimize impacts to the integrity of the system while capturing volatile forensic data

LIVE FORENSICS - ARTIFACTS

- System time
- Logged-on user(s)
- Open files
- Network information
- Network connections
- Process information
- Process-to-port mapping
- Process memory
- Network status
- Clipboard contents
- Service/driver information
- Command history
- Mapped drives
- Shares

FORENSIC VOLATILITY

- Volatile data: active information temporarily reflecting the machines current state including registers, caches, physical and virtual memory, network connections, shares, running processes, disks, floppy, tape, CD-ROM and printing activity.
- Generally, volatile data refers to, data residing in the RAM. This data is called “volatile” because it is lost the moment power supply goes off.
- For a forensic examiner these data are extremely important and they provide critical clues to different information relating to activities of a computer.
- For example, in an intrusion investigation, there may be a currently active network connection between the compromised system and a command and control host, or there may be a rootkit present only in memory, without an associated file on disk.

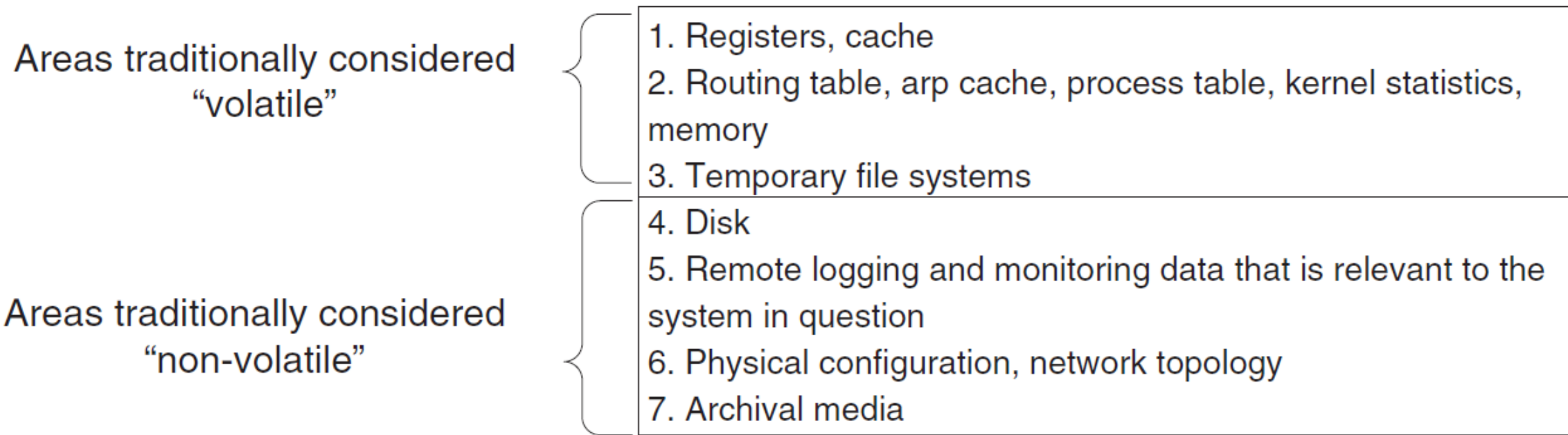
ORDER OF VOLATILITY

- The problem investigators face is the **order of volatility (OOV)**, which determines how long a piece of information lasts on a system.
- Memory
- Swap File
- Network Processes
- System Processes
- File System Information
- Raw Disk Blocks

ORDER OF VOLATILITY

RFC 3227 “Order of Volatility”

Areas traditionally considered
“volatile”

- 
1. Registers, cache
 2. Routing table, arp cache, process table, kernel statistics, memory
 3. Temporary file systems

Areas traditionally considered
“non-volatile”

4. Disk
5. Remote logging and monitoring data that is relevant to the system in question
6. Physical configuration, network topology
7. Archival media

SEARCH AND SEIZURE OF VOLATILE



WORKING PROCEDURE MANUAL COMPUTER FORENSICS



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Type of data	Life span
Registers, peripheral mem, cache, etc.	Nanoseconds
Main memory	Ten nanoseconds
Network state	Milliseconds
Running processes	Seconds
Disk	Minutes
Floppies, backup media, etc.	Years
CD-ROMs, DVDs, printouts, etc	Decades

GUIDELINES FOR COLLECTING VOLATILE DATA

- Avoid tools that use a GUI interface.
- Command line tools are best here
- Use safe and tested tools you know that work.
- Create two or three floppy disks containing your volatile collection tools and write protect them.
- Generate a checksum and validation for each of your tools and store it safely within your toolkit.

TOOLS FOR THE COLLECTION OF VOLATILE DATA

- Srvcheck.exe: A NTRK utility that displays the shares locally or remotely.
- Kill.exe: A Windows 2K Support tool for terminating a selected task or process.
- Rasusers.exe: A NTRK utility that lists all user accounts on a domain or server that have been granted permission to dial in to the network.
- Dumpel.exe: A NTRK utility to create an ASCII copy of the Event Viewer Logs.
- Filemon: A monitoring tool that displays all file system activity in real time.
- Regmon: A monitoring tool that displays all registry activity in real time.
- Tokenmon: A monitoring tool that displays logons, logoff, privilege usage and impersonation.

TOOLS FOR THE COLLECTION OF VOLATILE DATA

- Handle: A tool that displays what files are open by which processes and more.
- ListDLLs: A tool that lists all DLLs that are currently loaded including the version and the full path names of the loaded modules, etc.
- Process Explorer: A tool that displays open files, object processes, registry keys, DLLs and owners of object processes.
- MD5sum: A tool that generates the checksum of a file and provides verification.
- Fport: A tool that maps application processes to the ports they listen on.
- TCPView: A tool that shows the endpoints of all open TCP and UDP connections.
- Cmd.exe: The command prompt for Win NT/2000

REASONS TO COPY RAM

- Volatile memory is more trusted than non-volatile. This means, data that is usually stored encrypted in the hard drive, when loaded in memory will most likely not be protected. This includes passwords, financial transactions, encryption keys, etc.
- Malware can reside completely in memory and may not ever even touch the hard disk drive. This means that after removing the power from the target device, no record of the malware would exist.
- Memory is latent/dormant: Similar to recovery of deleted files, recovery of prior (deleted) processes is important in memory forensics. Eg: cached files may be stored in memory (and maybe never written to disk), thus making a RAM image more useful.

REASONS TO COPY RAM

- Evidence. Whether a malware was executed or not, can be proved using a memory image. This is because if the malware runs, it has to leave a footprint in the RAM
- Malware analysis. Executed code must exist somewhere in executable form, and sometimes it is better to analyze it from the RAM image. For instance, a packed executable are hard to understand; however, in some situations, the unpacked version of the binary could be retrieved directly from memory, making the malware analysis process much easier.

EXAMPLE SCENARIO: COMPUTER INTRUSION

- Operating system date and time: Critical for comparison to a central time source so that any variance in the time settings of the target system can be identified.
- List of running processes: Collected so that an examiner can later identify unauthorized or malicious processes that may have been active on the system.
- List of loaded drivers or modules: Collected to identify unauthorized or malicious code that may be loaded as a driver or module as opposed to a library or standard process.
- List of loaded libraries for each process: Collected to identify unauthorized or malicious code that may be running as a library loaded into an otherwise legitimate process.

EXAMPLE SCENARIO: COMPUTER INTRUSION

- List of open sockets and active network connections by process: Collected to identify any unauthorized communication sessions or open sockets that were active on the target system.
- Network configuration: Collected for various reasons, including the identification of anomalous configuration settings, as well as to simply understand the role of the system in the network in which it resides.
- List of file and Registry handles by process: Collected for various reasons, such as to make a determination as to what files and Registry entries may be connected to malicious processes.
- List of currently authenticated users: Collected to determine if there are any unauthorized authentications to the target system.

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